

PAVAN DHARMA ADAPA

Data Analyst · Business Intelligence

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PROFESSIONAL SUMMARY

Data analyst with an M.S. in Computer Science and a current analytics role in a federally funded research laboratory. Takes a question from raw source to a defensible answer — SQL and Python over real national microdata, Power BI and GA4 reporting for non-technical stakeholders, and a stated limitation on every result. Work-authorized through June 2027 on F-1 OPT and eligible for the 24-month STEM extension: no sponsorship required until 2029. Available immediately.

TECHNICAL SKILLS

Analysis & Programming: SQL (window functions, CTEs, dimensional queries), Python (pandas, NumPy, SciPy, scikit-learn), R, Advanced Excel, Jupyter

Reporting & BI: Power BI, Tableau, Google Analytics 4, stakeholder reporting, metric definition and standardization

Statistics & Data Handling: Hypothesis testing, power analysis, A/B testing and CUPED, survey-weighted estimation, forecasting (Random Forest, XGBoost), ETL and data cleaning, dimensional modeling, Git and continuous integration

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Rebuilt the lab's DNA sequencing pipeline into a documented, tested tool that installs in one command — **cutting setup from several days to under an hour** — across **34 merged pull requests** covering continuous integration, packaging, and cross-platform support.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus — the verification that makes the lab's published results defensible to reviewers.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**, and reported completion and grade metrics that let instructors reach at-risk students earlier.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Managed and analyzed a **900,000+ record** public-health surveillance dataset, and prepared the written and oral reports presented to faculty across two departments.
- Built the automated collection and cleaning workflow that standardized multi-source research data, **cutting per-dataset processing from hours to under 10 minutes** and making every figure reproducible from source.

Marketing Data Analyst Intern | *Viral Fission, Hyderabad, India* May 2023 – Dec 2023

- Built automated Power BI and Google Analytics 4 reporting for campaign performance, **halving manual reporting time** for a team running national brand campaigns.

ANALYTICS PROJECTS

Self-directed public projects — code, tests and continuous integration on GitHub; the data source is named on each.

Claims Denials & Appeals — *CMS Transparency in Coverage 2021–2023 · SQL, window functions, two-grain fact design*

- Modeled **15,545 plan-years** and **126.9M denied claims** across 624 issuer-years in a single fact design carrying two grains, using window functions for year-over-year comparison on real public issuer filings.
- Found only **0.168% of denials are ever appealed while 40.1% of appeals are overturned**, a rate that holds across all three years — locating the bottleneck in filing capacity rather than claim merit.

Experimentation Toolkit — *A/B test failure modes · simulation against known ground truth · Python, SciPy*

- Measured five A/B test failure modes against known ground truth rather than asserting them: peeking ten times at a null experiment drives the false-positive rate to **18.5% against a nominal 5%**, and CUPED at rho 0.7 cuts variance **49%**.
- Showed sample-ratio-mismatch detection depends entirely on scale — a 0.4% split imbalance is $p = 3e-141$ at 10M users and $p = 0.26$ at 20k — and validated every sample-size calculation by simulation: asked for 80% power, measured 80.1%.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.